

Centralizers of nilpotent wedges in real orthogonal Lie algebras

$\mathfrak{so}(p,q)$: a signature-resolved Levi-radical map

(Geometry of the vacuum, floor -1)

Volodymyr Sobol

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Abstract

For a rank-2 nilpotent (a wedge $N = x \wedge y$) in a real orthogonal Lie algebra $\mathfrak{so}(p,q)$, the centralizer $c(N)$ decomposes as a reductive part in semidirect product with a nilradical — a framework classical over algebraically closed fields (Springer–Steinberg; Collingwood–McGovern). We give the *signature-resolved* $\mathfrak{so}(p,q)$ form of this decomposition for the *enumerated* signatures $n = p+q \leq 6$: the non-abelian part is exhausted by four classes, each a 3-dimensional engine in semidirect product with a nilpotent module, falling into a dichotomy between the affine type $\mathfrak{so}(p',q') \ltimes \mathbb{R}^{\wedge \{d'\}}$ and the Jacobi type $\mathfrak{sp}(2m, \mathbb{R}) \ltimes \mathfrak{h}_{-}\{2m+1\}$. The governing mechanism is obtained for all n in block-symbolic form (by solving $[M,N] = 0$, not by fitting). A signature biconditional — *for the soft (rank-1) family*, the reductive Levi is compact iff $q = 1$ — follows from the definiteness of $(p-2, q-1)$. The explicit isomorphisms are machine-verified (Lean/mathlib, kernel-checked; axioms [propext, Classical.choice, Quot.sound], no native_decide). Three statements — the signature-resolved dichotomy table, the all- n centralizer form, and the soft-family compact-Levi biconditional — are absent from the literature surveyed (the nearest, Đoković–Lemire–Sekiguchi 2001, gives the closure order, not centralizer structure). No physical reading of any object is made or implied.

§1. Setup and framework

Let (V, η) be a real quadratic space of signature (p, q) , $n = p + q$, and $\mathfrak{so}(p, q)$ its orthogonal Lie algebra. A *wedge* $N = x \wedge y$ (rank-2 nilpotent, x, y isotropic and mutually orthogonal representatives) generates a nilpotent adjoint orbit; we study its centralizer $c(N) = \{A \in \mathfrak{so}(p,q) : [A, N] = 0\}$.

Over an algebraically closed field the structure $c(N) = (\text{reductive}) \ltimes (\text{unipotent radical})$ is classical (Springer–Steinberg 1970; Collingwood–McGovern, Thm 6.1.3), with reductive parts tabulated by partition for the complex classical algebras. Our contribution is the *real, signature-resolved* form: how the Levi $\ltimes$ radical splits as a function of (p, q) . The nearest classical real work, Đoković–Lemire–Sekiguchi 2001, classifies $O(p,q)$ -orbits and their closure order ($\min(p,q) \leq 7$) but not the centralizer structure — so the table below sits in a documented gap. (Explicit real centralizers are available for the exceptional algebras — Đoković 1988 — and Levi decompositions of nilpotent centralizers in classical groups are known over algebraically closed fields — Babinski–Stewart 2016 — but neither yields a signature-resolved $\mathfrak{so}(p,q)$ table.)

Scope (load-bearing qualifiers). All statements are for wedge nilpotents (rank $N = 2$); higher Jordan types are a named tail, out of scope. Strata are given by coordinate representatives; the invariants are representation-independent (checked on independent representatives of each

class). The classical frames invoked — Levi–Malcev decomposition, $\mathfrak{sl}(2)$ -module theory, the classification of Heisenberg algebras — are cited, not re-derived (multiplicity 0). Named objects that appear (Poincaré $2+1 = \mathfrak{iso}(2,1)$, Euclidean $\mathfrak{e}(3)$, oscillator/Jacobi $= \mathfrak{so}(2,1) \ltimes \mathfrak{h}_3$) are *flags* — they emerged by measurement on degeneracy strata, not by assumption.

§2. The signature-resolved centralizer map

Theorem (centralizer map, enumerated signatures $n \leq 6$). The non-abelian part of $c(N)$ for a wedge N in $\mathfrak{so}(p,q)$ is exhausted by four classes, each a 3-dimensional engine in semidirect product with a nilpotent module:

- **(i)** bare $\mathfrak{so}(2,1)$ — deep kernels $\ker = 2$, mixed J2/J4;
- **(ii)** $\text{Levi} \ltimes \mathbb{R}^3$ (irreducible 3): $\mathfrak{e}(3)$ iff $q = 1$, $\mathfrak{iso}(2,1)$ iff $q \geq 2$ — soft kernels $\ker = 4$;
- **(iii)** $\mathfrak{so}(2,1) \ltimes \mathfrak{h}_3$, with rad/Z the fundamental-2 (weights ± 1) — deep (3,2);
- **(iv)** $\mathfrak{so}(2,1) \ltimes \mathfrak{h}_5$, with rad/Z of type 2+2 (weights $\{+1, +1, -1, -1\}$, commutant-dim 4, min-cyclic 2; *not* the irreducible spin-3/2) — deep $n = 6$.

The solvable branch (no Levi factor) gives $(4,1)r1 \perp (3,2)r1$ as *distinct* algebras (the discriminant-form signature of ad on $[L, L]$: rotation $\pm i$ elliptic $\perp$ boost ± 1 hyperbolic). The abelian case: soft $\ker = 2$, and $(3,1)$ entirely (the centre is the whole centralizer).

Radical dichotomy (measured, no interpretation). Soft kernels carry an *abelian* (translational) radical; deep kernels carry a *Heisenberg* radical ($\mathfrak{h}_3/\mathfrak{h}_5$, canonical pairs in rad/Z , weights ± 1). Two species: translational $\perp$ symplectic.

Load-bearing qualifiers. This is an *enumeration of signatures* $n \leq 6$, not an arbitrary- n theorem (the mechanism derivation for all n is the named tail, closed at §3). "Engine compactness $\iff q = 1$ " holds *within the enumeration*. The classification is by structural invariants over the enumerated signatures; matching invariants are not claimed to pin the isomorphism class beyond the cases exhibited.

Address: probes S933, S935, S937, S940 (`src/symbolic/`, each with its reference log). *Citation anchor:* reductive $\times$ nilradical — Springer–Steinberg 1970, CM 6.1.3; the real $\mathfrak{so}(p,q)$ table is the white spot (nearest, DLS 2001, gives closures, not structure).

§3. The all- n centralizer form

Theorem (centralizer form, all n). The mechanism is derived for all n — not enumerated — by solving $[M, N] = 0$ in block-symbolic form (a matrix symbol with symbolic kernel dimension d); the resulting form is n -independent, obtained by *solving* the equations $\mathfrak{so}(\eta) \wedge [M, N] = 0$, not by fitting. In two families:

- **rank-0:** $c(N) \cong (\mathfrak{sp}(2, \mathbb{R}) \oplus \mathfrak{so}(\eta|G)) \ltimes \mathfrak{h}_{\lfloor 2d+1 \rfloor}$, with $d = n - 4$ and $\eta|G$ of signature $(p-2, q-2)$; the module is $F_2 \otimes V_{\lfloor d \rfloor}$, and *the centre is born from [module, module]* — the Heisenberg structure is not postulated, it falls out. $\dim c(N) = d(d-1)/2 + 2d + 4$.
- **rank-1:** $c(N) \cong \mathbb{R} \oplus (\mathfrak{so}(\eta|G') \ltimes \mathbb{R}^{\wedge \lfloor d' \rfloor})$, with $d' = n - 3$ and $\eta|G'$ of signature $(p-2, q-1)$; $\dim c(N) = d'(d'+1)/2 + 1$. Here *the Levi is compact iff $q = 1$* is now a **theorem** (definiteness of $(p-2, q-1)$ holds iff $q = 1$), not an empirical observation.

Load-bearing qualifiers. Only *wedge* nilpotents (rank $N = 2$; higher Jordan types are a named tail). The derivation is machine-symbolic (block_collapse) plus instance validation on $n = 5-7$ by span-equality (6/6) plus the closed form checked against all 15 measured centralizer dimensions (8/8 dimension classes), with the $n = 8$ control independently reproducing the rank-0 closed form at $d = 4$ (dim 18, h_9 , Levi $\mathfrak{so}(2,1) \oplus \mathfrak{so}(2,2)$; Lane-B visa J-0444); the orbit representative of a wedge is the Witt representative (cited, multiplicity 0). *A Lean-level formal proof is the endgame, not this act:* the free/linked/zero split of the block decomposition is *read* from the symbolic equations, not auto-solved end-to-end — the guard is the span-equality instance check on $n = 5-7$ (Lane-B visa J-0423).

Address: probe S952 (src/symbolic/, with its reference log). *Citation anchor:* the all- n centralizer form for $\mathfrak{so}(p,q)$ nilpotents is a white spot; its components (Levi decomposition, $\mathfrak{sl}(2)$ -modules, the Heisenberg parabolic — Jia, arXiv:2501.12406) are cited.

§4. The bridge: "construct = output"

For each core we *exhibit* an explicit two-sided isomorphism (bt_verify) between the centralizer core and the construct, given by explicit P/Q -aligned rational matrices (S946) — we do **not** appeal to an existence theorem (Levi–Malcev). Presenting the isomorphism, rather than invoking its existence, is exactly the machine-verifiable form used in §5.

Address: src/symbolic/S946_w33_leg2b_iso_export.json (6 isomorphisms, $n = 5, 6$); S946_w33_leg2b_bt_verify

§5. Machine verification

The explicit isomorphisms are verified in Lean 4 / mathlib (v4.32.0), as a fourth layer over the symbolic probes.

- **A1** — the (2,2) rank-0 core: centralizer of dimension 4 = $\mathfrak{so}(2,1) \oplus \mathbb{R}$ (centre); 16 theorems kernel-checked (So21.lean).
- **A2** — **the full centralizer $c(N)$, 6/6** (the $\mathbb{R}$ -centre for rank-1 and the reductive centre for rank-0 are included): So51r1 · So42r1 · So33r1 (rank-1, full dim 7 = core $\oplus \mathbb{R}$) and So32r0 · So42r0 · So33r0 (rank-0; the deep $n=6$ cases dim 9 = $(\mathfrak{sp}(2, \mathbb{R}) \oplus \mathfrak{so}(\eta|G)) \ltimes h_5$, the (3,2) case dim 6). 432 theorems, build clean.

Axiom audit. #printaxioms returns [propext, Classical.choice, Quot.sound] on every reported theorem — with **no sorry** and **no native_decide** (the tactic is decidable simp/norm_num; native_decide is deliberately avoided, being outside the trusted kernel via Lean.ofReduceBool).

Honest boundary. Lean proves {the basis is linearly independent, lies in $c(N)$, and closes with the measured bracket tables}; that the basis is *all* of $c(N)$ (dimension-completeness) is the symbolic probe (S933), not Lean.

Signature instance of the reductive centre $\mathfrak{so}(\eta|G) = \mathfrak{so}(p-2, q-2)$. For the deep $n=6$ rank-0 cores the reductive centre is the compact $\mathfrak{so}(2)$ or the non-compact boost $\mathfrak{so}(1,1)$ according to the signature: So42r0 (4,2) $\rightarrow$ the extra generator W satisfies $W^3 = -W$ (minimal polynomial $x(x^2+1)$, eigenvalues $\pm i$) $\rightarrow$ **compact $\mathfrak{so}(2)$** ((2,0) definite); So33r0 (3,3) $\rightarrow W^3 = +W$ ($x(x^2-1)$, eigenvalues ± 1) $\rightarrow$ **boost $\mathfrak{so}(1,1)$** ((1,1) indefinite) — at an *identical* core mechanism $\mathfrak{sp}(2, \mathbb{R}) \ltimes h_5$. Both $W \in \mathfrak{so}(\eta)$ and the eigenvalue-type invariant $W^3 = \mp W$ are named and kernel-checked (theorems B8_inS0, B8_cube). (Note: W^2 is *not* ± 1 on the full space — only on its 2-plane; the clean full-matrix invariant of the eigenvalue type is $W^3 = \mp W$.)

Address: Lean project in `lean/VGLearn/`; data `lean/artifacts/gen_A2_iso.py`, `src/symbolic/S946_w33_leg2`
Strongest card: machine verification renders our explicit real centralizers irrefutable — the literature gives the framework; we give verified instances in the white spot.

§6. Positioning: three documented gaps

The novelty rests on three statements absent from the literature we surveyed, each standing on a cited scaffold:

1. **A real $\mathfrak{so}(\mathfrak{p}, \mathfrak{q})$ Levi $\times$ radical table** (signature-resolved) — a gap; the nearest, Đoković–Lemire–Sekiguchi 2001, gives the closure order, not the structure.
2. **The affine $\iff$ Jacobi dichotomy** classifying wedge centralizers for $n \leq 6$ — a gap (a new packaging; the components — Eichler–Zagier; Berndt–Schmidt; Berceanu (explicit $\mathfrak{sp} \times$ Heisenberg commutators) — are cited).
3. **The "compact-Levi $\iff$ $\mathfrak{q} = 1$ " biconditional** — not written down anywhere as a theorem. It is *reinforced by machine instances* (§5): kernel-checked on both sides at an identical core — So42r0 compact $\mathfrak{so}(2) \perp$ So33r0 boost $\mathfrak{so}(1,1)$. We state "verified instances in a white spot," **not** "an $\forall$ -theorem kernel-checked" (the $\forall$ -biconditional is proved from signed-tableau definiteness — signed Young diagrams, Neuttiens–Pierard de Maujouy 2026; Lean supplies instances).

Reviewer risk. "The components are known" $\rightarrow$ the components, yes; but the dichotomy-as-classification for $\mathfrak{so}(\mathfrak{p}, \mathfrak{q})$ $n \leq 6$ and the signature biconditional are not written down; DLS 2001 does closures, not structure.

Card formulation (no over-claim). "The first explicit real **machine-verified instances** of the signature splitting of the reductive centre $\mathfrak{so}(\eta|G) = \mathfrak{so}(\mathfrak{p}-2, \mathfrak{q}-2)$; the general form is *enumerative*." This holds the boundary: the instances are kernel-checked $\perp$ the general $\forall$ -biconditional is signed-tableau (not to be re-claimed as "a theorem kernel-checked").

References

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Provenance (§0, per preprint-1): no result depends on a physical reading of any object, nor asserts one.